

Mobius 50 L - 2021 specification (NOT iFlex)

merck-mobius-2021-50

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	10 L	Source-supported
Maximum working volume	50 L	Source-supported
Nominal vessel / bag size	50 L	Source-supported
Total vessel volume loaded into the form	60 L	Source-supported
Initial working volume loaded into the form	10 L	Derived / estimated
Product-contact material	PureFlex bag film; polypropylene impeller	Source-supported
Height : diameter	1.62	Derived / estimated
Internal / chamber diameter	0.34 m	Source-supported
Liquid height at maximum working volume	0.5507 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	Source-supported
Impeller count	1	App assumption
Impeller diameter	0.1085 m	Source-supported
Power number	3.2	Source-supported
Initial agitation	50 rpm	App assumption
Maximum agitation	300 rpm	Source-supported
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	Yes	Source-supported
Maximum pressure	0.017 bar(g)	Source-supported
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 10 L

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 L.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Maximum working volume - 50 L

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 50 L.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Nominal vessel / bag size - 50 L

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 50 L.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Total vessel volume loaded into the form - 60 L

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 60 L.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Initial working volume loaded into the form - 10 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(10 L, 50 L) = 10 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - PureFlex bag film; polypropylene impeller

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 4; Materials of Construction, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: PureFlex bag film; polypropylene impeller.

Limitations: Bag film and main impeller material; 304L steel describes support hardware, not the product-contact bag. Other wetted components are listed separately in the original table.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 4](#)

Height : diameter - 1.62

Derived / estimated | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.05 \text{ m}^3$; $D = 0.34 \text{ m}$; $H = 0.5507091456466966 \text{ m}$; $H/D = 1.6197327813138134$

Limitations: Derived liquid aspect ratio. Does not reuse an overall vessel ratio or generic profile ratio as if it described the liquid.

Local original: Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2

Internal / chamber diameter - 0.34 m

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.34 m.

Limitations: The manufacturer row says Diameter; inside/outside convention is not explicit. Used as the model chamber diameter provisionally. This is not the external skid width on page 3.

Local original: Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2

Liquid height at maximum working volume - 0.5507 m

Derived / estimated | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.05 \text{ m}^3$; $D = 0.34 \text{ m}$; $H = 0.5507091456466966 \text{ m}$; $H/D = 1.6197327813138134$

Limitations: Ideal-cylinder estimate from published diameter and working volume. Ignores tapered walls, curved bottoms and internals; not a manufacturer liquid-height specification.

Local original: Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2

Impeller / mixing type - Pitched-blade turbine

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2

Impeller count - 1

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].impellerCount = 2`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.1085 m

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.1085 m.

Limitations: Absolute tabulated dimension converted from cm to m. Prefer this over the rounded 0.3 impeller/vessel ratio, which does not reproduce all listed dimensions.

Local original: Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2

Power number - 3.2

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 3.2.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Initial agitation - 50 rpm

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 50 rpm.

Limitations: Editable app starting point within the manufacturer operating range; not a published process setpoint.

Maximum agitation - 300 rpm

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 300 rpm.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Yes.

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Maximum pressure - 0.017 bar(g)

Source-supported | Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021) - PDF page 2; Specifications, Mobius 50 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.017 bar(g).

Limitations: SP1237EN00 Ver. 4.0, October 2021. Four blades at 13 degree pitch; bottom mounting 15 degrees from centre; X-baffle.

Maximum pressure converted from 17 mbar to 0.017 bar. These values do not apply to iFlex.

Local original: [Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 \(October 2021\) - PDF page 2](#)

Default operating pressure - 0 bar(g)

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0 bar(g).

Limitations: Atmospheric starting pressure selected by the app; below the documented 0.017 bar maximum.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

Separate 2021 Mobius model, not Mobius iFlex. User-supplied original archived unchanged; original publisher download URL was not provided.

Published agitation range: 30-300 rpm. Nominal total H:D is 2:1; no direct liquid height is specified. Liquid height and liquid H:D remain cylinder estimates.

Single-impeller count is a model interpretation of the bottom-mounted design. Starting gas rates and maximum vvm remain assumptions; individual filter/sparger flow limits are not summed into an unsupported total.

Page 3 equipment dimensions are external assembly dimensions, not liquid geometry. Jacket pressure is not the vessel operating limit.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Merck Mobius Bioreactors - SP1237EN00 Ver. 4.0 (October 2021)

merck-mobius-2021-sp1237en00-v4-3d185ccd3e36.pdf

SHA-256: 3d185ccd3e361a64bea5b97b3da620caf0be9183abb03b63a5f6837dc09f55ab